

Compound Poisson--Gamma, Tweedie Pure Premium, and Exposure Weights

Deriving the Tweedie pure-premium model from first principles - August 11, 2026

Overview

This note develops, from first principles, the connection between:

1. a Poisson model for claim counts
2. a Gamma model for individual claim severities
3. the compound Poisson–Gamma distribution for aggregate claim cost
4. the Tweedie distribution with power $1 < p < 2$
5. the pure premium (loss cost) as a rate per unit exposure
6. the appearance of `sample_weight = exposure` when fitting a Tweedie pure-premium model
7. the exact equivalence, in the Poisson case, between claim counts with a log-exposure offset and observed claim frequency with exposure as sample weight.

Overview

Definition 1: Central principle.

Exposure scales the **amount** of risk observed. If claim arrivals form a Poisson process, both the expected aggregate claim amount and its variance scale linearly with exposure.

The Tweedie result for pure premium is not introduced as an arbitrary distributional assumption. For $1 < p < 2$, it can be derived exactly from a compound Poisson model with Gamma severities.

1. Why notation matters

Two common uses of the symbol λ

A common source of confusion is the symbol λ .

In many texts,

$$N \sim \text{Poisson}(\lambda)$$

uses λ for the **Poisson mean**, i.e. the expected number of events over the observation interval.

In insurance texts one also often sees

$$N_i \sim \text{Poisson}(w_i \lambda_i)$$

where λ_i is now interpreted as a **claim-frequency rate per unit exposure**.

Both conventions are valid, but using the same symbol for an absolute expected count and for a rate easily obscures the role of exposure.

This note deliberately uses two different symbols.

Core notation

1.1 Notation

For policy or observation i :

Symbol	Meaning	Typical unit
X_i	rating variables / covariates	–
$e_i > 0$	exposure	policy-years
ν_i	claim-frequency (a rate)	claims / policy-year
Λ_i	expected claim count over observed exposure	claims
N_i	observed number of claims	claims
$Z_{\text{sub } \{ij\}}$	amount of claim j	EUR / claim
ζ_i	expected severity	EUR / claim
S_i	observed aggregate claim amount	EUR

Core notation

Symbol	Meaning	Typical unit
μ_i	expected pure-premium rate	EUR / policy-year
M_i	expected aggregate claim amount	EUR
R_i	observed pure premium $\frac{S_i}{e_i}$	EUR / policy-year
p	Tweedie power	dimensionless
φ	base Tweedie dispersion parameter	model-dependent units

The key identities are

$$\Lambda_i = e_i \nu_i$$

and pure premium = claim frequency x severity:

$$\mu_i = \nu_i \zeta_i$$

Consequently: $M_i = e_i \mu_i$

2. Frequency: rate versus expected count

Frequency: rate versus expected count

Assume claim arrivals follow a Poisson process conditional on the policy characteristics X_i . Let

$$\nu_i = \nu(X_i)$$

denote the claim-frequency rate. If exposure is measured in policy-years, then ν_i has units

$$\frac{\text{claims}}{\text{policy-year}}$$

The expected number of claims during exposure e_i is

$$\Lambda_i = e_i \nu_i$$

We model

$$N_i \mid X_i, e_i \sim \text{Poisson}(\Lambda_i) = \text{Poisson}(e_i \nu_i)$$

Frequency: rate versus expected count

Thus,

$$\mathbb{E}[N_i \mid X_i, e_i] = e_i \nu_i$$

and because a Poisson random variable has variance equal to its mean,

$$\text{Var}(N_i \mid X_i, e_i) = e_i \nu_i$$

Concrete numerical illustration

Suppose a fixed frequency $\nu_i = 0.10$ “claims per policy-year”.

For a full year,

$$e_i = 1 \Rightarrow \Lambda_i = 0.10$$

For half a year,

$$e_i = 0.5 \Rightarrow \Lambda_i = 0.05$$

For three months,

$$e_i = 0.25 \Rightarrow \Lambda_i = 0.025$$

Nothing changes about the annualized rate.

Exposure only converts that rate into the expected count over the actually observed risk period. This is what phrases such as “unit exposure” mean: one simply evaluates the process at $e_i = 1$.

3. Severity model

Gamma severity model

Let $Z_{i1}, Z_{i2}, \dots$ be individual claim amounts for policy i . Conditional on X_i , assume the claim amounts are iid and independent of N_i . Write

$$\zeta_i = \mathbb{E}[Z_{ij} \mid X_i]$$

for mean severity.

For the exact Tweedie compound-Poisson representation we assume Gamma severities. A Gamma distribution is a reasonable assumption, having a strictly positive support and a not too thin tail.

To avoid ambiguity between Gamma **scale** and **rate** conventions, this note uses a shape-rate parameterization:

$$Z_{ij} \mid X_i \sim \text{Gamma}(\gamma, c_i)$$

where $\gamma > 0$ is the shape and $c_i > 0$ is the rate. Therefore,

Gamma severity model

$$\mathbb{E}[Z_{ij} \mid X_i] = \frac{\gamma}{c_i} = \zeta_i$$

$$\text{Var}(Z_{ij} \mid X_i) = \frac{\gamma}{c_i^2}$$

$$\mathbb{E}[Z_{ij}^2 \mid X_i] = \frac{\gamma(\gamma + 1)}{c_i^2}$$

4. Aggregate claim cost

Compound Poisson aggregate loss

Define aggregate claim cost

$$S_i = \sum_{j=1}^{N_i} Z_{ij}$$

with the usual convention that $S_i = 0$ when $N_i = 0$. We have

$$N_i \mid X_i, e_i \sim \text{Poisson}(e_i \nu_i)$$

and iid Gamma severities. Therefore S_i has a compound Poisson–Gamma distribution. This is the classical collective risk model underlying the Tweedie distribution for $1 < p < 2$; see Jørgensen and de Souza [2] and Delong, Lindholm and Wüthrich [1].

5. Mean aggregate claim cost

Mean aggregate claim cost

Condition first on the number of claims. Given N_i ,

$$\mathbb{E}[S_i \mid N_i, X_i, e_i] = N_i \zeta_i$$

Taking expectation again,

$$\mathbb{E}[S_i \mid X_i, e_i] = \mathbb{E}[N_i \mid X_i, e_i] \zeta_i = e_i \nu_i \zeta_i$$

Define the expected pure-premium rate

$$\mu_i = \nu_i \zeta_i$$

Hence

$$\mathbb{E}[S_i \mid X_i, e_i] = e_i \mu_i$$

This is the actuarial identity

$$\text{pure premium rate} = \text{claim-frequency rate} \times \text{mean severity}$$

6. Variance of aggregate claim cost

Law of total variance

Using the law of total variance,

$$\text{Var}(S_i \mid X_i, e_i) = \mathbb{E}[\text{Var}(S_i \mid N_i, X_i, e_i) \mid X_i, e_i] + \text{Var}(\mathbb{E}[S_i \mid N_i, X_i, e_i] \mid X_i, e_i)$$

Given N_i ,

$$\text{Var}(S_i \mid N_i, X_i, e_i) = N_i \text{Var}(Z_{ij} \mid X_i)$$

so the first term is $\mathbb{E}[N_i \mid X_i, e_i] \text{Var}(Z_{ij} \mid X_i)$. The second term is $\zeta_i^2 \text{Var}(N_i \mid X_i, e_i)$.

Because N_i is Poisson,

$$\mathbb{E}[N_i \mid X_i, e_i] = \text{Var}(N_i \mid X_i, e_i) = e_i \nu_i$$

Therefore

$$\text{Var}(S_i \mid X_i, e_i) = e_i \nu_i [\text{Var}(Z_{ij} \mid X_i) + \zeta_i^2]$$

General compound Poisson variance identity

But

$$\text{Var}(Z_{ij}) + (\mathbb{E}[Z_{ij}])^2 = \mathbb{E}[Z_{ij}^2]$$

Hence

$$\text{Var}(S_i \mid X_i, e_i) = e_i \nu_i \mathbb{E}[Z_{ij}^2 \mid X_i]$$

This result is true for a general compound Poisson model; Gamma severity has not yet been needed nor assumed.

Definition 2: Linear exposure scaling.

The exposure dependence is already explicit: $\text{Var}(S_i \mid X_i, e_i) \propto e_i$. This linear scaling follows from the Poisson-process assumption.

7. The exact Tweedie connection

Distributional equivalence

For $1 < p < 2$, the Tweedie distribution is exactly a compound Poisson distribution with Gamma jumps.

This is not merely a similarity of first two moments. The equivalence is **distributional**.

Delong, Lindholm and Wüthrich [1, Sections 2.1–2.2, Proposition 2.1] give a particularly clear derivation by comparing moment-generating functions. They start from a compound Poisson–Gamma model with claim count mean equal to exposure times claim frequency and identify the corresponding Tweedie parameters.

Jørgensen and de Souza [2] is a foundational actuarial reference for the same compound-Poisson Tweedie process.

8. Mean-dispersion notation for the Tweedie distribution

Tweedie parameterization

To keep the derivation transparent, define

$$Y \sim \text{Tw}_p(m, d)$$

to mean a Tweedie random variable satisfying

$$\mathbb{E}[Y] = m \quad \text{and} \quad \text{Var}(Y) = dm^p$$

Here m is the mean, d is the actual dispersion attached to this observation, and $p \in (1, 2)$ is the Tweedie power.

Later, for a pure-premium observation with exposure e_i , we will obtain

$$d_i = \frac{\varphi}{e_i}$$

This notation deliberately separates the **observation-specific dispersion** d_i from the common base parameter φ .

9. Mapping Tweedie parameters to the compound Poisson–Gamma model

Parameter correspondence

For $Y \sim \text{Tw}_p(\mu, \varphi)$ with $1 < p < 2$, the associated compound Poisson–Gamma representation can be written as $Y = \sum_{j=1}^N Z_j$ where the Poisson intensity is

$$\nu = \frac{\mu^{2-p}}{\varphi(2-p)}$$

and the Gamma severities have shape

$$\gamma = \frac{2-p}{p-1}$$

and rate

$$c = \frac{\mu^{1-p}}{\varphi(p-1)}$$

Parameter correspondence

These identities are equivalent to the parameter correspondence in Delong, Lindholm and Wüthrich [1, Proposition 2.1].

10. Recovering frequency times severity

Recovering the pure-premium rate

The mean Gamma severity is $\zeta = \frac{\gamma}{c}$. Substituting the Tweedie parameter mapping,

$$\zeta = \frac{\frac{2-p}{p-1}}{\frac{\mu^{1-p}}{\varphi(p-1)}} = \varphi(2-p)\mu^{p-1}$$

The claim-frequency rate is

$$\nu = \frac{\mu^{2-p}}{\varphi(2-p)}$$

Multiplying frequency and severity,

$$\nu\zeta = \frac{\mu^{2-p}}{\varphi(2-p)} \times \varphi(2-p)\mu^{p-1} = \mu^{2-p+p-1} = \mu$$

Hence

Recovering the pure-premium rate

$$\mu = \nu\zeta$$

Thus the Tweedie mean is precisely the actuarial pure-premium rate.

11. Explicit derivation of the Tweedie variance rate

Variance rate derivation (part 1)

We now show explicitly why $\nu \mathbb{E}[Z^2] = \varphi \mu^p$.

For a Gamma variable with shape γ and rate c ,

$$\mathbb{E}[Z^2] = \frac{\gamma(\gamma + 1)}{c^2}$$

From $\gamma = \frac{2-p}{p-1}$ we get

$$\gamma + 1 = \frac{2-p}{p-1} + 1 = \frac{1}{p-1}$$

Therefore

$$\gamma(\gamma + 1) = \frac{2-p}{(p-1)^2}$$

Variance rate derivation (part 2)

Also,

$$c = \frac{\mu^{1-p}}{\varphi(p-1)} \quad \text{so} \quad c^2 = \frac{\mu^{2-2p}}{\varphi^2(p-1)^2}$$

Hence

$$\mathbb{E}[Z^2] = (2-p)\varphi^2\mu^{2p-2}$$

Now multiply by the frequency rate $\nu = \frac{\mu^{2-p}}{\varphi(2-p)}$:

$$\nu \mathbb{E}[Z^2] = \frac{\mu^{2-p}}{\varphi(2-p)} \times (2-p)\varphi^2\mu^{2p-2} = \varphi\mu^p$$

The exponent simplifies to $2-p+2p-2=p$. Therefore

$$\nu \mathbb{E}[Z^2] = \varphi\mu^p$$

Variance rate derivation (part 2)

This is the exact origin of the Tweedie variance function in the compound Poisson–Gamma representation.

12. Introducing arbitrary exposure

Exposure scaling of mean and variance

Return to policy i with exposure e_i . The underlying claim-frequency rate is ν_i . The Poisson mean count over the actual observed exposure is

$$\Lambda_i = e_i \nu_i$$

Thus $N_i \sim \text{Poisson}(e_i \nu_i)$ and aggregate loss is $S_i = \sum_{j=1}^{N_i} Z_{ij}$.

From the general compound Poisson result,

$$\text{Var}(S_i \mid X_i, e_i) = e_i \nu_i \mathbb{E}[Z_{ij}^2 \mid X_i]$$

From the Tweedie parameter mapping just proved,

$$\nu_i \mathbb{E}[Z_{ij}^2 \mid X_i] = \varphi \mu_i^p$$

Therefore

$$\text{Var}(S_i \mid X_i, e_i) = e_i \varphi \mu_i^p$$

Exposure scaling of mean and variance

Together with $\mathbb{E}[S_i \mid X_i, e_i] = e_i \mu_i$, both mean and variance scale **linearly** with risk exposure.

13. From aggregate loss to observed pure premium

Pure premium and its variance

The observed pure premium (also called loss cost or loss rate) is

$$R_i = \frac{S_i}{e_i}$$

Its conditional mean is

$$\mathbb{E}[R_i \mid X_i, e_i] = \mu_i$$

Its conditional variance is

$$\text{Var}(R_i \mid X_i, e_i) = \frac{1}{e_i^2} \text{Var}(S_i \mid X_i, e_i) = \frac{e_i \varphi \mu_i^p}{e_i^2}$$

Hence

$$\text{Var}(R_i \mid X_i, e_i) = \frac{\varphi}{e_i} \mu_i^p$$

Pure premium and its variance

This already identifies the exposure weight.

14. Exact distribution of the pure premium

Demonstration by scaling

$$R_i = \frac{S_i}{e_i} = \sum_{j=1}^{N_i} \frac{Z_{ij}}{e_i}$$

The count remains $N_i \sim \text{Poisson}(e_i \nu_i)$ and

$$\frac{Z_{ij}}{e_i} \sim \text{Gamma}(\gamma, e_i c_i)$$

Thus R_i is again compound Poisson–Gamma, with:

- Poisson mean = $e_i \nu_i$
- Gamma shape = γ
- Gamma rate = $e_i c_i$

Because compound Poisson–Gamma distributions are exactly Tweedie for $1 < p < 2$, R_i is Tweedie.

Demonstration by matching parameters

Suppose $R_i \sim \text{Tw}_p(\mu_i, d_i)$. The compound Poisson intensity implied by this Tweedie distribution is $\frac{\mu_i^{2-p}}{d_i(2-p)}$. Set $d_i = \frac{\varphi}{e_i}$. Then the implied Poisson intensity is

$$\frac{\mu_i^{2-p}}{\left(\frac{\varphi}{e_i}\right)(2-p)} = e_i \frac{\mu_i^{2-p}}{\varphi(2-p)} = e_i \nu_i = \Lambda_i$$

The Tweedie-implied Gamma rate is also $e_i c_i$. The Gamma shape remains $\gamma = \frac{2-p}{p-1}$. All parameters match exactly, so

$$R_i \mid X_i, e_i \sim \text{Tw}_p\left(\mu_i, \frac{\varphi}{e_i}\right)$$

15. The aggregate loss is Tweedie too

Scaling property of Tweedie

The Tweedie family has the scaling property

$$Y \sim \text{Tw}_p(m, d) \Rightarrow aY \sim \text{Tw}_p(am, a^{2-p}d)$$

Set $a = e_i$. Then $S_i = e_i R_i$ has dispersion $e_i^{2-p} \frac{\varphi}{e_i} = \varphi e_i^{1-p}$.

Therefore

$$S_i \mid X_i, e_i \sim \text{Tw}_p(e_i \mu_i, \varphi e_i^{1-p})$$

Check the variance:

$$\text{Var}(S_i) = \varphi e_i^{1-p} (e_i \mu_i)^p = \varphi e_i^{1-p} e_i^p \mu_i^p = e_i \varphi \mu_i^p$$

This agrees exactly with the compound Poisson calculation.

16. Why exposure is the sample weight

Weight derivation

The pure-premium response satisfies $R_i \sim \text{Tw}_p\left(\mu_i, \frac{\varphi}{e_i}\right)$, which is equivalent to

$$\text{Var}(R_i \mid X_i, e_i) = \frac{\varphi \mu_i^p}{e_i}$$

In an exponential dispersion model, an observation weight w_i is represented through $\text{Var}(Y_i \mid X_i) = \frac{\varphi V(\mu_i)}{w_i}$. With $V(\mu_i) = \mu_i^p$ for Tweedie, comparing the two expressions gives

$$w_i = e_i$$

Therefore, for an actuarial pure-premium model,

$$y_i = R_i = \frac{S_i}{e_i}, \quad \text{sample weight} = e_i$$

This is the statistical justification for

Weight derivation

```
y_pure_premium = total_claim_amount / exposure

model.fit(
    X,
    y_pure_premium,
    sample_weight=exposure,
)
```

when the model is `TweedieRegressor` or a weighted Tweedie objective such as LightGBM's Tweedie regression. The current scikit-learn insurance example uses exactly this pure-premium construction and exposure weighting [6].

Practical implementation

scikit-learn

```
from sklearn.linear_model \
    import TweedieRegressor

y = total_claim_amount /
exposure

model = TweedieRegressor(
    power=p,
    link="log",
    alpha=0.0,
)

model.fit(
    X, y,
```

LightGBM

```
import lightgbm as lgb

y = total_claim_amount /
exposure

model = lgb.LGBMRegressor(
    objective="tweedie",
    tweedie_variance_power=p,
)

model.fit(
    X, y,
    sample_weight=exposure,
)
```

Practical implementation

```
    sample_weight=exposure,  
)
```

For actuarial interpretation: $\hat{\mu}_i$ is the predicted pure-premium rate and $\hat{M}_i = e_i \hat{\mu}_i$ is the predicted aggregate loss.

17. The same result from the Tweedie likelihood

Likelihood weighting

The variance derivation is sufficient for the quasi-likelihood argument: quasi-likelihood specifies the conditional mean and variance function, rather than a full conditional distribution. Hence, once the exposure scaling of the variance is specified, the corresponding rate weight follows directly from

$$\text{Var}(R_i \mid x_i) = \frac{\varphi}{e_i} \mu_i^p.$$

For a full Tweedie exponential dispersion model, we can also see the weight directly in the likelihood. Write the Tweedie density schematically as

$$f\left(r_i; \theta_i, \frac{\varphi}{e_i}, p\right) = \exp\left\{ \frac{r_i \theta_i - \kappa_p(\theta_i)}{\frac{\varphi}{e_i}} + a\left(r_i, \frac{\varphi}{e_i}, p\right) \right\}$$

Since $\frac{1}{\frac{\varphi}{e_i}} = \frac{e_i}{\varphi}$, the part of the log-likelihood depending on the mean parameter is $\frac{e_i}{\varphi} [r_i \theta_i - \kappa_p(\theta_i)]$.

Each observation contributes proportionally to e_i , confirming exposure is a **precision / likelihood weight**.

Likelihood weighting

DeLong, Lindholm and Wüthrich [1, equations (2.2)–(2.5)] formulate the Tweedie exponential dispersion family directly with exposure w and obtain $\text{Var}(Y) = \frac{\varphi}{w} \mu^p$. Our notation corresponds to $w = e_i$.

18. The same result from Tweedie deviance

Tweedie deviance identity

For $p \neq 1, 2$, the Tweedie unit deviance is

$$d_p(y, m) = 2 \left[\frac{y^{2-p}}{(1-p)(2-p)} - \frac{ym^{1-p}}{1-p} + \frac{m^{2-p}}{2-p} \right]$$

The pure-premium likelihood with dispersion $\frac{\varphi}{e_i}$ leads, for estimation of the mean model, to an objective proportional to

$$\sum_i e_i d_p(R_i, \mu_i)$$

Thus the actuarial pure-premium objective is

$$\sum_i e_i d_p\left(\frac{S_i}{e_i}, \mu_i\right)$$

This is exactly the form implemented when the pure-premium response is fitted with `sample_weight = e_i`.

19. A common misconception

Common dispersion on aggregate loss

A common conceptual mistake is to write $S_i \sim \text{Tw}_p(e_i \mu_i, \varphi)$ with the **same aggregate dispersion φ for every exposure**. That model implies

$$\text{Var}(S_i) = \varphi(e_i \mu_i)^p = \varphi e_i^p \mu_i^p$$

which is not linear in exposure for $p \neq 1$.

By contrast, the compound Poisson process gives $\text{Var}(S_i) = e_i \varphi \mu_i^p$, so the aggregate Tweedie dispersion must be $\varphi_{S,i} = \varphi e_i^{1-p}$, not a common constant.

Definition 3: Correct pure-premium model.

$R_i \sim \text{Tw}_p\left(\mu_i, \frac{\varphi}{e_i}\right)$, hence `sample_weight = e_i`.

**20. What a single Tweedie model
assumes about frequency and severity**

Frequency-severity link

There is an important restriction hidden inside a single Tweedie pure-premium model with fixed p and common base dispersion φ . The compound-Poisson representation gives

$$\nu_i = \frac{\mu_i^{2-p}}{\varphi(2-p)} \quad \text{and} \quad \zeta_i = \varphi(2-p)\mu_i^{p-1}$$

Thus frequency and mean severity are both functions of the same μ_i ; they satisfy $\mu_i = \nu_i \zeta_i$ but cannot vary completely independently if p and φ are fixed.

This is one reason actuarial practice often fits separate Poisson frequency and Gamma severity GLMs. See Delong, Lindholm and Wüthrich [1] for the detailed relationship, and Smyth and Jørgensen [5] for dispersion modelling when a homogeneous Tweedie dispersion is too restrictive.

21. Modeling interpretation

Frequency-severity link

Five-step summary

Definition 4: The five modeling steps.

1. **Claim arrivals:** $N_i \mid X_i, e_i \sim \text{Poisson}(e_i \nu_i)$. Here ν_i is a claim-frequency rate.
2. **Claim severity:** $Z_{ij} \mid X_i \sim \text{Gamma}(\gamma, c_i)$. $\zeta_i = \frac{\gamma}{c_i}$.
3. **Aggregate cost:** $S_i = \sum_{j=1}^{N_i} Z_{ij}$. $\mathbb{E}[S_i] = e_i \nu_i \zeta_i = e_i \mu_i$.
4. **Pure premium:** $R_i = \frac{S_i}{e_i}$. $\mathbb{E}[R_i] = \mu_i = \nu_i \zeta_i$.
5. **Tweedie representation:** $R_i \sim \text{Tw}_p\left(\mu_i, \frac{\varphi}{e_i}\right)$, hence `sample_weight` = e_i .

22. Minimal practical implementations

scikit-learn

```
from sklearn.linear_model import TweedieRegressor

y = total_claim_amount / exposure

model = TweedieRegressor(
    power=p,
    link="log",
    alpha=0.0,
)

model.fit(
    X,
    y,
    sample_weight=exposure,
)
```


scikit-learn

For actuarial interpretation: $\hat{\mu}_i$ is the predicted pure premium per unit exposure, and $\widehat{M}_i = e_i \hat{\mu}_i$ is the predicted aggregate loss.

LightGBM

```
import lightgbm as lgb

y = total_claim_amount / exposure

model = lgb.LGBMRegressor(
    objective="tweedie",
    tweedie_variance_power=p,
)

model.fit(
    X,
    y,
    sample_weight=exposure,
)
```

Again, $\hat{\mu}_i$ is the predicted pure-premium rate and $\hat{M}_i = e_i \hat{\mu}_i$ is the predicted aggregate loss.

23. Assumptions behind exposure weighting

Modelling assumptions

The result $\text{Var}(R_i) = \frac{\varphi \mu_i^p}{e_i}$ follows from a stochastic exposure model. The basic assumptions include:

1. exposure measures claim-generating risk volume
2. conditional claim-arrival intensity is proportional to exposure
3. the risk is sufficiently homogeneous over the exposure interval
4. severity does not mechanically change because the observation window is shorter
5. claim arrivals satisfy the Poisson assumption (the working frequency model)
6. for the exact Tweedie result, severities are Gamma
7. frequency and severity are conditionally independent under the collective-risk construction

These assumptions can be questionable under strong seasonality, endogenous cancellation, catastrophe dependence, or within-policy coverage changes.

24. Summary of the main derivation

Derivation summary (part 1)

Start with a frequency rate ν_i , convert to expected count $\Lambda_i = e_i \nu_i$, model $N_i \sim \text{Poisson}(\Lambda_i)$, let mean severity be ζ_i , so pure premium is $\mu_i = \nu_i \zeta_i$.

Aggregate loss is $S_i = \sum_{j=1}^{N_i} Z_{ij}$.

For compound Poisson: $\text{Var}(S_i) = e_i \nu_i \mathbb{E}[Z_i^2]$.

For the Tweedie parameterization: $\nu_i \mathbb{E}[Z_i^2] = \varphi \mu_i^p$.

Therefore $\text{Var}(S_i) = e_i \varphi \mu_i^p$.

Derivation summary (part 2)

Define observed pure premium $R_i = \frac{S_i}{e_i}$. Then $\mathbb{E}[R_i] = \mu_i$ and $\text{Var}(R_i) = \varphi \frac{\mu_i^p}{e_i}$.

Under the exact Gamma-severity assumption, $R_i \sim \text{Tw}_p\left(\mu_i, \frac{\varphi}{e_i}\right)$, hence the actuarial Tweedie pure-premium fit is $y_i = \frac{S_i}{e_i}$ with `sample_weight = e_i`.

Appendix A. Poisson counts with exposure offset versus claim- frequency rates

A.1 Notation

This appendix proves the exact analogue for claim frequency. The result is simpler than for Tweedie because Poisson corresponds to the boundary power $p = 1$.

For policy i :

- e_i : exposure
- ν_i : expected claim-frequency rate
- Λ_i : expected claim count over the observed exposure
- N_i : observed claim count
- $F_i = \frac{N_i}{e_i}$: observed claim-frequency rate

The connection is $\Lambda_i = e_i \nu_i$.

Assume a log-link frequency model $\log \nu_i = X_i^\top \beta$, so $\nu_i = \exp(X_i^\top \beta)$.

A.2 Count with a log-exposure offset

The count model is $N_i \sim \text{Poisson}(\Lambda_i)$ with $\Lambda_i = e_i \nu_i$. Taking logs,

$$\log \Lambda_i = \log e_i + X_i^\top \beta$$

Thus $\log e_i$ is an **offset**: its coefficient is fixed at one.

The Poisson log-likelihood for observation i is

$$\ell_i = N_i \log(e_i \nu_i) - e_i \nu_i - \log(N_i!)$$

Expanding, the terms $N_i \log e_i - \log(N_i!)$ do not depend on β . Therefore, for estimating β ,

$$\ell_i(\beta) \equiv N_i \log \nu_i - e_i \nu_i$$

Using $N_i = e_i F_i$, this becomes $\ell_i(\beta) \equiv e_i [F_i \log \nu_i - \nu_i]$ – already the weighted-rate formulation.

A.3 Rate with exposure weight

Define the observed claim-frequency rate $F_i = \frac{N_i}{e_i}$.

Its conditional expectation is $\mathbb{E}[F_i \mid X_i, e_i] = \nu_i$ and its variance is

$$\text{Var}(F_i \mid X_i, e_i) = \frac{\nu_i}{e_i}$$

A weighted Poisson EDF / quasi-likelihood model has $\text{Var}(F_i) = \frac{\nu_i}{w_i}$. Thus $w_i = e_i$.

Lemma 1: Caveat.

$F_i = \frac{N_i}{e_i}$ is generally not an integer and should not be interpreted literally as a Poisson count. The rate formulation is an exponential-dispersion / quasi-likelihood representation whose estimating equations and deviance are equivalent to those of the count-with-offset formulation.

A.4 Equality of the score equations

For the count formulation, $\ell_i(\beta) \equiv N_i \log \nu_i - e_i \nu_i$. Under the log link, $\partial \frac{\nu_i}{\partial} \beta = \nu_i X_i$, so the score is $\partial \frac{\ell_i}{\partial} \beta = X_i(N_i - e_i \nu_i)$.

Summing over observations,

$$U_{\text{count}}(\beta) = \sum_i X_i(N_i - e_i \nu_i)$$

For the rate formulation with weight e_i ,

$$U_{\text{rate}}(\beta) = \sum_i e_i X_i(F_i - \nu_i) = \sum_i X_i(N_i - e_i \nu_i)$$

Hence $U_{\text{rate}}(\beta) = U_{\text{count}}(\beta)$. The two unpenalized estimators solve exactly the same score equation.

A.5 Poisson deviance identity

The Poisson unit deviance is

$$d_P(y, m) = 2 \left[y \log \left(\frac{y}{m} \right) - (y - m) \right]$$

with the convention $0 \log 0 = 0$. One verifies directly that

$$d_P(N_i, e_i \nu_i) = e_i d_P \left(\frac{N_i}{e_i}, \nu_i \right)$$

Summing over observations,

$$\text{Poisson count} + \text{offset} \log e_i \Leftrightarrow \text{rate } \frac{N_i}{e_i} + \text{weight } e_i$$

for the unpenalized Poisson deviance / likelihood mean model.

A.6 Why Poisson is especially simple

Boundary case

For Tweedie power $p = 1$,

$$e_i^{2-p} = e_i^1 = e_i$$

Several exposure-scaling identities that differ for $1 < p < 2$ collapse to the same expression in the Poisson boundary case. This is one reason the offset-versus-rate equivalence for claim frequency is less controversial and easier to see.

A.7 Practical formulations

Count + offset

```
response = claim_count
mean      = exposure *
frequency_rate
link      = log
offset    = log(exposure)
```

Rate + exposure weight

```
response      =
claim_count / exposure
sample_weight = exposure
link          = log
```

$$\log \mathbb{E}[N_i] = \log e_i + X_i^\top \beta$$

$$\log \mathbb{E}[F_i] = X_i^\top \beta$$

For an unpenalized Poisson mean model, the two formulations produce the same estimating equations. With regularization or sample-weight normalization, one should check the exact implemented objective before claiming identical numerical fits.

Appendix B. Compact notation map to Delong, Lindholm and Wüthrich

Notation map

Delong, Lindholm and Wüthrich [1] use $N \sim \text{Poi}(\lambda w)$, where w is exposure and their λ is the expected claim-frequency rate relative to exposure.

This note renames those quantities as $w \leftrightarrow e$, $\lambda \leftrightarrow \nu$ and introduces separately

$$\Lambda = e\nu$$

for the absolute Poisson mean count over the observed exposure.

They define aggregate claims $S = \sum_{j=1}^N Z_j$ and show that the exposure-normalized quantity $\frac{S}{w}$ has the Tweedie exponential-dispersion representation with $\mathbb{E}\left[\frac{S}{w}\right] = \mu$ and $\text{Var}\left(\frac{S}{w}\right) = \frac{\varphi}{w}\mu^p$.

This is exactly our result $R_i = \frac{S_i}{e_i} \sim \text{Tw}_p\left(\mu_i, \frac{\varphi}{e_i}\right)$.

References

Core actuarial references

1. [1] Delong, L., Lindholm, M., & Wüthrich, M. V. (2021). **Making Tweedie's compound Poisson model more accessible**. European Actuarial Journal, **11**, 185–226. doi: [10.1007/s13385-021-00264-3](https://doi.org/10.1007/s13385-021-00264-3).
2. [2] Jørgensen, B., & de Souza, M. C. P. (1994). **Fitting Tweedie's compound Poisson model to insurance claims data**. Scandinavian Actuarial Journal, **1994**(1), 69–93. doi: [10.1080/03461238.1994.10413930](https://doi.org/10.1080/03461238.1994.10413930).
3. [3] Wüthrich, M. V. **Non-Life Insurance: Mathematics & Statistics**. SSRN Manuscript ID 2319328, living lecture notes.
4. [4] Ohlsson, E., & Johansson, B. (2010). **Non-Life Insurance Pricing with Generalized Linear Models**. Springer.
5. [5] Smyth, G. K., & Jørgensen, B. (2002). **Fitting Tweedie's compound Poisson model to insurance claims data: dispersion modelling**. ASTIN Bulletin, **32**(1), 143–157. doi: [10.2143/AST.32.1.1020](https://doi.org/10.2143/AST.32.1.1020).

Software reference

1. [6] scikit-learn developers. **Tweedie regression on insurance claims**, official scikit-learn example.

Final takeaway

Software reference

For standard non-life ratemaking, the Tweedie pure-premium fit $y_i = \frac{S_i}{e_i}$ with `sample_weight = e_i` is induced exactly by the compound Poisson–Gamma construction.